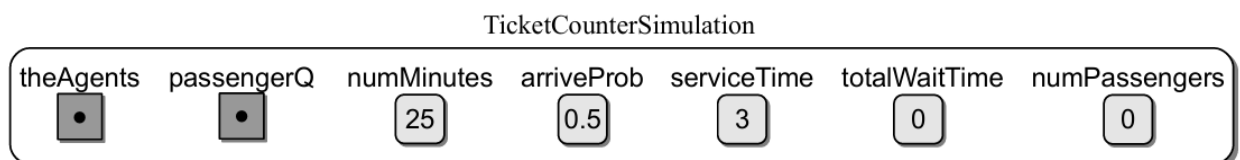


```

1  # Implementation of the main simulation class.
2  from array import Array
3  from llistqueue import Queue
4  from people import TicketAgent, Passenger
5
6  class TicketCounterSimulation :
7      # Create a simulation object.
8      def __init__( self, numAgents, numMinutes, betweenTime, serviceTime ) :
9          # Parameters supplied by the user.
10         self._arriveProb = 1.0 / betweenTime
11         self._serviceTime = serviceTime
12         self._numMinutes = numMinutes
13
14         # Simulation components.
15         self._passengerQ = Queue()
16         self._theAgents = Array( numAgents )
17         for i in range( numAgents ) :
18             self._theAgents[i] = TicketAgent(i+1)
19
20         # Computed during the simulation.
21         self._totalWaitTime = 0
22         self._numPassengers = 0
23
24         # Run the simulation using the parameters supplied earlier.
25     def run( self ) :
26         for curTime in range(self._numMinutes + 1) :
27             self._handleArrival( curTime )
28             self._handleBeginService( curTime )
29             self._handleEndService( curTime )
30
31         # Print the simulation results.
32     def printResults( self ) :
33         numServed = self._numPassengers - len(self._passengerQ)
34         avgWait = float( self._totalWaitTime ) / numServed
35         print( "" )
36         print( "Number of passengers served = ", numServed )
37         print( "Number of passengers remaining in line = %d" %
38             len(self._passengerQ) )
39         print( "The average wait time was %4.2f minutes." % avgWait )
40
41     # The remaining methods that have yet to be implemented.
42     # def _handleArrive( curTime ) :      # Handles simulation rule #1.
43     # def _handleBeginService( curTime ) : # Handles simulation rule #2.
44     # def _handleEndService( curTime ) :   # Handles simulation rule #3.

```



Gambar 1. Contoh TicketCounterSimulationobject.

1. Tentukan kompleksitas waktu (worst case) untuk setiap operasi yang didefinisikan dalam **class TicketCounterSimulation**

Num Minutes	Num Agents	Average Service	Time Between	Average Wait	Passengers Served	Passengers Remaining
100	2	3	2	2.49	49	2
500	2	3	2	3.91	240	0
1000	2	3	2	10.93	490	14
5000	2	3	2	15.75	2459	6
10000	2	3	2	21.17	4930	18
100	2	4	2	10.60	40	11
500	2	4	2	49.99	200	40
1000	2	4	2	95.72	400	104
5000	2	4	2	475.91	2000	465
10000	2	4	2	949.61	4000	948
100	3	4	2	0.51	51	0
500	3	4	2	0.50	240	0
1000	3	4	2	1.06	501	3
5000	3	4	2	1.14	2465	0
10000	3	4	2	1.21	4948	0

Table 8.1: Sample results of the ticket counter simulation experiment.

2. Lakukan eksekusi manual kode berikut dan tampilkan isi queue yang dihasilkan:

```
values = Queue()
for i in range(16):
    if i % 3 == 0:
        values.enqueue(i)
```
3. Lakukan eksekusi manual kode berikut dan tampilkan isi queue yang dihasilkan:

```
values = Queue()
for i in range(16):
    if i % 3 == 0:
        values.enqueue(i)
    elif i % 4 == 0:
        values.dequeue()
```
4. Implementasikan metode yang tersisa dari *class TicketCounterSimulation*.
5. Modifikasi kelas TicketCounterSimulation agar menggunakan satuan detik, bukan menit. Jalankan eksperimen dan buat tabel seperti Tabel 8.1.
6. Rancang dan implementasikan fungsi untuk membalik urutan item dalam queue. Hanya boleh menggunakan operasi Queue ADT (boleh pakai struktur data lain).